MSG No. 211.57 $I4321'$ [Type II, cubic]

Table 1: Wyckoff site: 2a, site symmetry: 4321'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,
		73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96]

Table 2: Wyckoff site: 6b, site symmetry: 42.21'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,8,9,10,13,14,49,50,51,56,57,58,61,62]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[4,5,15,16,18,19,21,23,52,53,63,64,66,67,69,71]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[6,7,11,12,17,20,22,24,54,55,59,60,65,68,70,72]
4	$\left[\frac{1}{2}, 0, 0\right]$	[25,26,27,32,33,34,37,38,73,74,75,80,81,82,85,86]
5	$\left[0, 0, \frac{1}{2}\right]$	[28,29,39,40,42,43,45,47,76,77,87,88,90,91,93,95]
6	$\left[0, \frac{1}{2}, 0\right]$	[30,31,35,36,41,44,46,48,78,79,83,84,89,92,94,96]

Table 3: Wyckoff site: 8c, site symmetry: .321'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,17,18,36,38,40,49,65,66,84,86,88]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,7,15,33,43,48,50,55,63,81,91,96]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,4,11,34,44,45,51,52,59,82,92,93]
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5,6,13,32,46,47,53,54,61,80,94,95]
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[8,22,23,29,30,37,56,70,71,77,78,85]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[9,19,24,26,31,39,57,67,72,74,79,87]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[10,20,21,27,28,35,58,68,69,75,76,83]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[12,14,16,25,41,42,60,62,64,73,89,90]

Table 4: Wyckoff site: 12d, site symmetry: 2.221'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[1,8,37,38,49,56,85,86]
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[2,3,33,34,50,51,81,82]
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[4,16,42,45,52,64,90,93]
4	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[5,15,43,47,53,63,91,95]
5	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[6,11,44,46,54,59,92,94]
6	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[7,12,41,48,55,60,89,96]
7	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[9,10,26,27,57,58,74,75]
8	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[13,14,25,32,61,62,73,80]
9	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[17,24,31,36,65,72,79,84]
10	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[18,21,28,40,66,69,76,88]
11	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[19,23,29,39,67,71,77,87]

continued ...

Table 4

No.	position	mapping
12	$\left[0,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[20,22,30,35,68,70,78,83]`

Table 5: Wyckoff site: `12e`, site symmetry: `4..1'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2,3,8,49,50,51,56]`
2	$[0,\ 0,\ -x]$	`[4,16,19,23,52,64,67,71]`
3	$[0,\ 0,\ x]$	`[5,15,18,21,53,63,66,69]`
4	$[0,\ x,\ 0]$	`[6,11,17,24,54,59,65,72]`
5	$[0,\ -x,\ 0]$	`[7,12,20,22,55,60,68,70]`
6	$[-x,\ 0,\ 0]$	`[9,10,13,14,57,58,61,62]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[25,26,27,32,73,74,75,80]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[28,40,43,47,76,88,91,95]`
9	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[29,39,42,45,77,87,90,93]`
10	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[30,35,41,48,78,83,89,96]`
11	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[31,36,44,46,79,84,92,94]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[33,34,37,38,81,82,85,86]`

Table 6: Wyckoff site: `16f`, site symmetry: `.3.1'`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,17,18,49,65,66]`
2	$[x,\ -x,\ x]$	`[2,7,15,50,55,63]`
3	$[x,\ x,\ -x]$	`[3,4,11,51,52,59]`
4	$[-x,\ x,\ x]$	`[5,6,13,53,54,61]`
5	$[x,\ -x,\ -x]$	`[8,22,23,56,70,71]`
6	$[-x,\ x,\ -x]$	`[9,19,24,57,67,72]`
7	$[-x,\ -x,\ x]$	`[10,20,21,58,68,69]`
8	$[-x,\ -x,\ -x]$	`[12,14,16,60,62,64]`
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[25,41,42,73,89,90]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[26,31,39,74,79,87]`
11	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[27,28,35,75,76,83]`
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[29,30,37,77,78,85]`
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[32,46,47,80,94,95]`
14	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[33,43,48,81,91,96]`
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[34,44,45,82,92,93]`
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[36,38,40,84,86,88]`

Table 7: Wyckoff site: `24g`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,8,49,56]`
2	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[2,3,50,51]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\ \frac{1}{2},\ -x\right]$	[4,16,52,64]
4	$\left[0,\ \frac{1}{2},\ x\right]$	[5,15,53,63]
5	$\left[\frac{1}{2},\ x,\ 0\right]$	[6,11,54,59]
6	$\left[\frac{1}{2},\ -x,\ 0\right]$	[7,12,55,60]
7	$\left[-x,\ \frac{1}{2},\ 0\right]$	[9,10,57,58]
8	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[13,14,61,62]
9	$\left[0,\ x,\ \frac{1}{2}\right]$	[17,24,65,72]
10	$\left[\frac{1}{2},\ 0,\ x\right]$	[18,21,66,69]
11	$\left[\frac{1}{2},\ 0,\ -x\right]$	[19,23,67,71]
12	$\left[0,\ -x,\ \frac{1}{2}\right]$	[20,22,68,70]
13	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[25,32,73,80]
14	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[26,27,74,75]
15	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[28,40,76,88]
16	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[29,39,77,87]
17	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[30,35,78,83]
18	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[31,36,79,84]
19	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[33,34,81,82]
20	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[37,38,85,86]
21	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[41,48,89,96]
22	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[42,45,90,93]
23	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[43,47,91,95]
24	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[44,46,92,94]

Table 8: Wyckoff site: 24h, site symmetry: ..21'

No.	position	mapping
1	$[0,\ y,\ y]$	[1,13,49,61]
2	$[0,\ -y,\ y]$	[2,10,50,58]
3	$[0,\ y,\ -y]$	[3,9,51,57]
4	$[y,\ y,\ 0]$	[4,18,52,66]
5	$[-y,\ y,\ 0]$	[5,19,53,67]
6	$[-y,\ 0,\ y]$	[6,20,54,68]
7	$[y,\ 0,\ y]$	[7,17,55,65]
8	$[0,\ -y,\ -y]$	[8,14,56,62]
9	$[y,\ 0,\ -y]$	[11,22,59,70]
10	$[-y,\ 0,\ -y]$	[12,24,60,72]
11	$[y,\ -y,\ 0]$	[15,23,63,71]
12	$[-y,\ -y,\ 0]$	[16,21,64,69]
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ y+\frac{1}{2}\right]$	[25,37,73,85]
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	[26,34,74,82]
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	[27,33,75,81]
16	$\left[y+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[28,42,76,90]
17	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[29,43,77,91]
18	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[30,44,78,92]
19	$\left[y+\frac{1}{2},\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[31,41,79,89]

continued ...

Table 8

No.	position	mapping
20	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$	[32,38,80,86]
21	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[35,46,83,94]
22	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[36,48,84,96]
23	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[39,47,87,95]
24	$\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[40,45,88,93]

Table 9: Wyckoff site: 24i, site symmetry: ..21'

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	[1,38,49,86]
2	$\left[\frac{1}{4},\ y+\frac{1}{2},\ y\right]$	[2,33,50,81]
3	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ -y\right]$	[3,34,51,82]
4	$\left[\frac{1}{2}-y,\ y,\ \frac{3}{4}\right]$	[4,45,52,93]
5	$\left[y+\frac{1}{2},\ y,\ \frac{1}{4}\right]$	[5,47,53,95]
6	$\left[-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[6,46,54,94]
7	$\left[y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[7,48,55,96]
8	$\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$	[8,37,56,85]
9	$\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$	[9,26,57,74]
10	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$	[10,27,58,75]
11	$\left[y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[11,44,59,92]
12	$\left[-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[12,41,60,89]
13	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ y\right]$	[13,32,61,80]
14	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -y\right]$	[14,25,62,73]
15	$\left[\frac{1}{2}-y,\ -y,\ \frac{1}{4}\right]$	[15,43,63,91]
16	$\left[y+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	[16,42,64,90]
17	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	[17,36,65,84]
18	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[18,40,66,88]
19	$\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[19,39,67,87]
20	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$	[20,35,68,83]
21	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[21,28,69,76]
22	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$	[22,30,70,78]
23	$\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[23,29,71,77]
24	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	[24,31,72,79]

Table 10: Wyckoff site: 48j, site symmetry: 11'

No.	position	mapping
1	$[x,\ y,\ z]$	[1,49]
2	$[x,\ -z,\ y]$	[2,50]
3	$[x,\ z,\ -y]$	[3,51]
4	$[z,\ y,\ -x]$	[4,52]
5	$[-z,\ y,\ x]$	[5,53]
6	$[-y,\ x,\ z]$	[6,54]
7	$[y,\ -x,\ z]$	[7,55]

continued ...

Table 10

No.	position	mapping
8	$[x, -y, -z]$	[8,56]
9	$[-x, y, -z]$	[9,57]
10	$[-x, -y, z]$	[10,58]
11	$[y, x, -z]$	[11,59]
12	$[-y, -x, -z]$	[12,60]
13	$[-x, z, y]$	[13,61]
14	$[-x, -z, -y]$	[14,62]
15	$[z, -y, x]$	[15,63]
16	$[-z, -y, -x]$	[16,64]
17	$[z, x, y]$	[17,65]
18	$[y, z, x]$	[18,66]
19	$[-y, z, -x]$	[19,67]
20	$[-z, -x, y]$	[20,68]
21	$[-y, -z, x]$	[21,69]
22	$[z, -x, -y]$	[22,70]
23	$[y, -z, -x]$	[23,71]
24	$[-z, x, -y]$	[24,72]
25	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[25,73]
26	$\left[x+\frac{1}{2}, \frac{1}{2}-z, y+\frac{1}{2}\right]$	[26,74]
27	$\left[x+\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-y\right]$	[27,75]
28	$\left[z+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-x\right]$	[28,76]
29	$\left[\frac{1}{2}-z, y+\frac{1}{2}, x+\frac{1}{2}\right]$	[29,77]
30	$\left[\frac{1}{2}-y, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[30,78]
31	$\left[y+\frac{1}{2}, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[31,79]
32	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[32,80]
33	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[33,81]
34	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[34,82]
35	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-z\right]$	[35,83]
36	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[36,84]
37	$\left[\frac{1}{2}-x, z+\frac{1}{2}, y+\frac{1}{2}\right]$	[37,85]
38	$\left[\frac{1}{2}-x, \frac{1}{2}-z, \frac{1}{2}-y\right]$	[38,86]
39	$\left[z+\frac{1}{2}, \frac{1}{2}-y, x+\frac{1}{2}\right]$	[39,87]
40	$\left[\frac{1}{2}-z, \frac{1}{2}-y, \frac{1}{2}-x\right]$	[40,88]
41	$\left[z+\frac{1}{2}, x+\frac{1}{2}, y+\frac{1}{2}\right]$	[41,89]
42	$\left[y+\frac{1}{2}, z+\frac{1}{2}, x+\frac{1}{2}\right]$	[42,90]
43	$\left[\frac{1}{2}-y, z+\frac{1}{2}, \frac{1}{2}-x\right]$	[43,91]
44	$\left[\frac{1}{2}-z, \frac{1}{2}-x, y+\frac{1}{2}\right]$	[44,92]
45	$\left[\frac{1}{2}-y, \frac{1}{2}-z, x+\frac{1}{2}\right]$	[45,93]
46	$\left[z+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-y\right]$	[46,94]
47	$\left[y+\frac{1}{2}, \frac{1}{2}-z, \frac{1}{2}-x\right]$	[47,95]
48	$\left[\frac{1}{2}-z, x+\frac{1}{2}, \frac{1}{2}-y\right]$	[48,96]